



**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2011
Higher 2**

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CANDIDATE NAME

CLASS

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INDEX NUMBER

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BIOLOGY

9648/03

Paper 3

20 September 2011

Additional Materials: Answer Booklet/ Paper

2 Hours

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX NO. and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your answers to the structured questions and essay separately.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Total	

This paper consists of **14** printed pages including the cover page

Answer all questions

- 1 Corn (or maize) is one of the most important agricultural crops as it acts as a staple in the diet of livestock and humans alike. To combat a certain species of seed corn beetles, genetic engineering was used to produce a variant of corn known as the *Bt* corn. **Figure 1.1** shows the sequence of events in the production of *Bt* corn.

rn cells

Figure 1.1

- (a) (i)** State **two** reasons why farmers would prefer to grow *Bt* corn.

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..... [2]

- (ii)** With reference to Figure 1.1, explain how *Bt* gene from the bacteria is isolated and then inserted into corn cells.

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..... [3]